Chapter 11 Enhancing Engineering Education: A Multimodal Approach to Personalization and Adaptation Using Artificial Intelligence in

Game-Based Learning

Roger Azevedo[1], Daryn Dever[2], and Megan Wiedbusch[1]

[1] School of Modeling, Simulation, and Training, University of Central Florida, FL, USA

[2] College of Education, University of Florida, FL, USA

11.1 Context: Challenges in Engineering Education

Engineering education presents a formidable challenge for many students due to a combination of factors that make the field inherently complex and demanding. At its core, engineering involves the application of complex concepts including mathematics, physics, and other sciences, requiring students to develop a deep understanding of abstract principles and apply them to real-world problems (National Academy of Engineering 2018). This level of abstract thinking and problem-solving intensity can be particularly challenging for those who are more comfortable with concrete concepts or struggle with spatial reasoning (National Academies of Sciences, Engineering, and Medicine 2018).

Engineering program curricula are typically rigorous and fast-paced, with heavy course loads that address a wide range of topics. Students often find themselves juggling with multiple challenging courses, projects, and laboratory work simultaneously, which demands excellent self-regulatory skills such as time management skills and the ability to quickly assimilate and apply new information (Chen et al. 2021). This rapid pace of learning, combined with high internal and external academic expectations, can create a stressful environment for many students, especially those who may need more time to fully grasp complex concepts (Geisinger and Raman 2013).

Another significant challenge in engineering education is the disconnect that some students experience between theoretical concepts and their practical applications. In the early years of study, it can be difficult for students to see how the abstract principles they learn relate to real-world engineering problems (Pawley 2009). This lack of context can make it harder for some students to stay motivated or fully engage with the material. Additionally, engineering often requires proficiency in various technical skills, including the use of specialized software tools and programming languages, which adds another layer of complexity to the learning process (Kober 2015).

The collaborative nature of many engineering projects can also present challenges for some students (Chen et al. 2021). While teamwork is an essential skill in the professional world, students who prefer individual work or struggle with interpersonal skills may find group projects particularly demanding. Moreover, the field of engineering is constantly evolving, with new technologies such as AI and methodologies emerging regularly. This rapid pace of change requires students to be adaptable and willing to continuously update their knowledge and skills (National Academy of Engineering 2013). The current engineering curriculum is designed to instill this adaptability and the importance of continuous learning, ensuring that students are well prepared for the dynamic nature of the engineering industry.

Lastly, the analytical thinking required in engineering education can be a significant hurdle for some students. Engineering problems often have multiple possible solutions, requiring students to critically analyze various approaches and make informed decisions based on complex criteria (Dym et al. 2005). Developing this level of analytical skill takes time, practice, and feedback and can be frustrating for students who are used to more straightforward problem-solving methods with simple solutions.

This is to say that a serious combination of challenging spatial and reasoning concepts, intense workloads, lack of participatory or active learning pedagogy often results in a loss of student interest or field abandonment (Ruiperez-Valiente et al. 2017). However, with the right support systems in place such as personalized support for the enactment of self-regu-

latory skills embedded within interactive engaging technological learning environments, students can navigate these challenges more effectively. GBLEs in particular may be uniquely situated to address these challenges, by easing the complexity of certain engineering subjects or concepts and encouraging retention through increased interest through interactivity and immersion (Butler and Ahmed 2016).

Despite these challenges, many students find engineering education to be rewarding and intellectually stimulating. Success in this field often requires dedication, perseverance, and the development of effective study strategies (Bernold et al. 2007). As engineering educators continue to refine their teaching methods and incorporate new technologies such as game-based learning, simulations, immersive virtual environments, and digital twins of mechanical and manufacturing systems, into the curriculum, there is an ongoing effort to make engineering education more accessible and engaging for a more diverse population of students. This commitment to improvement should inspire hope and optimism for the future of engineering education.

In this chapter, we argue that the challenges experienced by students in engineering courses are significant and urgent and can be effectively addressed using learning sciences and SRL principles (e.g. Azevedo and Dever 2022; Azevedo and Wiedbusch 2023; Sawyer 2023; Winne and Azevedo 2023). By focusing on the development, use, and transfer of students' SRL strategies involving metacognitive monitoring and regulation of their cognitive, affective, metacognitive, and motivational processes (Azevedo et al. 2018; Winne 2018; Winne and Azevedo 2022), we can pave the way for more engaging and effective learning experiences. Moreover, we propose using GBLEs to significantly enhance students' engagement and interactivity while fostering a deep understanding of complex engineering concepts and phenomena (Dever et al. 2023a, 2024). This approach, coupled with the collection of students' SRL process data from various sensors during game-based learning, can provide real-time, adaptive scaffolding and feedback to meet each student's SRL and learning needs using various AI-based techniques from the field of AI in education (see du Boulay et al. 2023). Our approach is based on principles of adap-

tivity and personalization (Azevedo et al. 2024; Plass and Pawar 2020) that will be described in further detail in the following sections.

11.2 GBLEs for Engineering Education: Are they Effective?

GBLEs have been instrumental in addressing the aforementioned engineering challenges by providing interactive simulations and scenarios that contextualize abstract concepts and theories in practical applications (Plass et al. 2019). For example, engineering students can engage in virtual laboratories or simulations to manipulate variables and observe outcomes, reinforcing their understanding of complex causal principles through experiential learning (de Jong et al. 2023). These environments offer immediate feedback and opportunities for iterative problem-solving, which is essential for developing proficiency in challenging engineering topics.

For example, GBLEs such as those used in electrical engineering to teach circuit design and analysis have demonstrated their potential to enhance the learning process. These games allow students to build and test virtual circuits, providing immediate feedback and reducing the fear of failure associated with physical lab experiments (Coller and Scott 2009). This approach has shown promise in not only improving students' engagement but also enhancing their understanding of abstract electrical concepts. Similarly, GBLEs have been employed in mechanical engineering to teach principles of dynamics and control systems. These interactive environments allow students to manipulate variables and observe the effects in real time, enhancing their understanding of cause-and-effect relationships in complex mechanical systems (Deshpande and Huang 2011).

As such, GBLEs can enhance engagement and motivation by presenting these abstract concepts and theories in a visually compelling and interactive manner on demand that helps sustain students' interest in otherwise complex and de-personalized educational environments (Mayer 2019). By integrating gamification elements such as scoring systems, achievements, and levels, educators can foster a competitive spirit and a sense of

achievement among students, further enhancing their learning experience (Rahimi et al. 2022).

While GBLEs have made significant strides in addressing some of the challenges in engineering education, several issues remain. One persistent challenge is the development of critical thinking, problem-solving, and SRL skills. While games can provide engaging contexts for learning, translating these skills to open-ended, real-world engineering problems remains difficult. The structured nature of most games, with clear objectives and rules, may not fully prepare students for the ambiguity and complexity of real-world engineering problems. Moreover, the immediate feedback provided in GBLEs may not always be available in real-world situations, making it challenging for students to develop the patience and persistence required for SRL (Ke 2009).

The assessment of the learning outcomes of GBLEs presents ongoing challenges, underscoring the complexities involved in evaluating learning in these interactive settings. Traditional assessment methods may not adequately capture the complex, multifaceted learning that occurs in these interactive settings, necessitating the development of new evaluation approaches, such as stealth assessment (Shute 2023) based on the collection of students' multimodal trace data during gameplay (Dever et al. 2022, 2023b, 2024; Taub et al. 2019), to address each student's learning needs.

In conclusion, while GBLEs have shown promise in addressing some challenges in engineering education, particularly in visualizing complex systems and bridging the theory-practice gap, significant work remains. The future of engineering education hinges on the continued research and development of GBLEs. This should focus on enhancing students' self-regulatory skills, providing more sophisticated assessment methods to diagnose students' deep conceptual knowledge (and not just factual or problem-solving steps), and adapting and personalizing learning materials, instructional methods, problem sets, based on the capabilities of GBLEs in detecting, measuring, modeling, tracking, fostering, and supporting each learner's SRL needs during gameplay.

11.3 Personalization and Adaptivity in GBLEs

Personalization and adaptivity are two related but distinct concepts that aim to enhance the learning experience in GBLEs (Ninaus and Nebel 2021). *Personalization* in game-based learning refers to tailoring game content, interface features, or learning paths based on individual learner characteristics, preferences, skills and abilities, or prior knowledge. This customization is typically preset or user-controlled and aims to create a more engaging and relevant learning experience for each user to avoid unnecessary repetition of mastered materials or the frustration of being pushed too quickly through challenging content they need more time to understand. We operationally define *personalization* as the degree to which a GBLE allows for or automatically implements changes in content presentation, difficulty level, or interface based on predetermined user characteristics or explicit user choices (Vandewaetere et al. 2011; Walkington and Bernacki 2020).

Similarly, *adaptivity* in game-based learning is a dynamic process involving the automatic adjustment of the learning experience based on the learner's real-time performance, behavior, or physiological responses within the game. This process, typically system-controlled, aims to optimize the learning experience by maintaining an appropriate level of challenge and support within an individual. We define *adaptivity* as the extent to which a GBLE autonomously modifies its instructional approach, content difficulty, or feedback mechanisms in real-time, based on continuous assessment of learner performance, engagement metrics, or physiological data (Hwang et al. 2017; Lopes and Bidarra 2011). While personalization and adaptivity are distinct concepts, they often overlap in practice. Many GBLEs incorporate personalization and adaptivity features to create more effective and engaging learning experiences (see Mayer 2019; Plass et al. 2019).

11.4 Personalization and Adaptivity in GBLEs for Engineering Education: Are they Effective?

Personalization and adaptivity, as critical features in GBLEs, can significantly enhance the effectiveness of educational games by tailoring the

learning experience to individual learners (Lamb et al. 2018). The empirical research on personalization and adaptivity in game-based learning generally indicates positive effects on learning, reasoning, and problem-solving across various domains and educational contexts (Clark et al. 2016). With effect sizes ranging from small to large, and most falling in the moderate range, these approaches show promise.

A meta-analysis by Vlachopoulos and Makri (2017) examined the effectiveness of games and simulations in higher education, including engineering disciplines. They reported an overall positive effect of game-based learning on learning outcomes, with a moderate effect size of d = 0.67. While this meta-analysis was not exclusive to personalized or adaptive games, it provides a baseline for understanding the general effectiveness of game-based approaches in higher education settings, including engineering.

Focusing more specifically on personalization in engineering education, a synthesis by Kickmeier-Rust and Albert (2010) investigated the effects of a personalized educational game for teaching basic electrical engineering concepts. They reported a significant improvement in learning outcomes with a large effect size (d = 0.89) when comparing the personalized game to a nonpersonalized version. This suggests that incorporating personalization features in engineering games can substantially enhance learning effectiveness.

Adaptivity in game-based learning for engineering has also shown promising results. A study by Hwang et al. (2020) examined the effectiveness of an adaptive game-based learning approach for teaching programming concepts to engineering students. They found that the adaptive system led to significantly better learning outcomes than a nonadaptive version, with a large effect size (d = 0.82). This indicates that adaptive features can improve learning outcomes in complex engineering topics such as programming.

Regarding problem-solving skills, which are crucial in engineering education, a study by Gong et al. (2019) investigated the effects of an adaptive GBLE on engineering students' problem-solving abilities. They reported a moderate to large effect size (d = 0.76) for improvement in problem-solv-

ing skills when comparing the adaptive game-based approach to traditional instruction methods. This suggests that adaptive game-based learning can be particularly effective in developing essential problem-solving competencies in engineering students.

The effectiveness of personalization and adaptivity in game-based learning for engineering also extends to specific engineering domains. For instance, in mechanical engineering, a study by Coller and Scott (2009) investigated using a personalized video game to teach numerical methods. They reported a large effect size (d = 0.87) for learning gains when comparing the game-based approach to traditional instruction. While this study did not explicitly measure the effect of personalization, it demonstrates the potential of tailored game-based approaches in specific engineering subjects.

In civil engineering, Ebner and Holzinger (2007) studied the effectiveness of an adaptive educational game for teaching structural concrete design. They found a moderate effect size (d = 0.58) for learning outcomes when comparing the adaptive game to traditional teaching methods. This suggests that adaptive game-based approaches can effectively teach complex engineering concepts that require spatial reasoning and structural analysis skills.

It is important to note that the effectiveness of personalization and adaptivity in game-based learning for engineering can vary depending on the specific aspects being personalized or adapted. Tseng et al. (2020) examined different types of adaptivity in a GBLE for engineering students. They found that adapting the difficulty level of challenges based on student performance had a more significant effect (d = 0.71) on learning outcomes than adapting the content presentation (d = 0.45). This highlights the importance of considering which aspects of the game to personalize or adapt for maximum effectiveness.

The impact of personalization and adaptivity on student motivation in engineering game-based learning has also been studied. A meta-analysis by Sailer and Homner (2020) on gamification in education, which included studies on engineering, found a small to moderate effect size (g = 0.47) for the impact of gamification on motivation. While not exclusively focused

on personalization and adaptivity, their analysis suggests that game elements, which often include personalized features, can positively influence student motivation in engineering education.

Across these meta-analyses, the effectiveness of these approaches appears to be influenced by factors such as the type of adaptivity implemented, the nature of the learning task, the educational level of the learners, and the specific domain of study (Boyle et al. 2016; Tsai and Tsai 2020). In conclusion, empirical research on personalization and adaptivity in game-based learning for engineering generally indicates positive effects on learning, reasoning, and problem-solving. Effect sizes range from moderate to large, with most falling in the moderate to high range. The effectiveness of these approaches is influenced by factors such as the specific engineering domain, the type of skills being developed, and the aspects of the game being personalized or adapted. While the results are promising, they also underscore the need for more research to fully understand the optimal design and implementation of personalized and adaptive features in game-based learning for engineering education (Koskinen et al. 2023).

We argue that it is not only important but urgent to conduct further research on personalization and adaptivity in GBLEs in engineering education by leveraging AI solutions. By concentrating on students' SRL processes during gameplay using multimodal trace data, we can identify and monitor the development, use, and transfer of various SRL processes. These data can then be utilized in near or real time to personalize and adapt to each student's learning and SRL needs.

11.5 Augmenting Personalization and Adaptivity in GBLEs in Engineering Education with Multimodal Trace Data

Multimodal multichannel data refers to the multiple data types and data streams that can be collected to triangulate covert psychological processes that occur during learning, especially within technology-enhanced environments, such as GBLEs (Azevedo and Gasevic 2019). This data encompasses a wide range of modalities, including but not limited to click-

stream data, eye-tracking information, physiological measures, natural language inputs, screen recordings of learner–system interactions, and facial expressions of emotions (Azevedo 2015; Azevedo et al. 2013, 2018; Azevedo and Gasevic 2019; D'Mello et al. 2017). Multimodal multichannel data traces provide a granular view of learners' cognitive, affective, metacognitive, and motivational processes as they unfold in real time, capturing both overt behaviors and covert processes and enabling researchers to triangulate information from different sources, providing a richer, more nuanced picture of learners' self-regulatory activities (Taub et al. 2019).

These data can significantly enhance personalization and adaptivity in GBLEs for engineering education by providing a more comprehensive understanding of learners' SRL processes (Azevedo et al. 2018; Dever et al. in press; Wiedbusch et al. 2023). Moreover, they can be leveraged to create more responsive and tailored learning experiences, ultimately improving learning outcomes and engagement in engineering education. However, despite the informative nature of rich multimodal traces, the analysis and utilization of such data often requires sophisticated and resource intensive computational techniques, including machine learning (ML) and data mining approaches, to uncover meaningful patterns and relationships (Azevedo and Gasevic 2019; Giannakos et al. 2022; Kovanović et al. 2023). This analysis can inform the development of GBLEs and personalized interventions to support SRL (Roll and Winne 2015). By capturing the complexity and dynamism of SRL processes, multimodal trace data offers unprecedented opportunities for understanding and fostering effective learning strategies in technology-enhanced environments (Azevedo 2015; Molenaar et al. 2023) and can be used strategically, including the case of personalization and adaptivity in GBLEs in engineering education.

One of the primary ways multimodal data can enhance personalization is by integrating various types of physiological measures and affective computing techniques with nuanced behavioral measures in addition to more traditional self-reported or performance based measures. *Physiological measures*, such as eye-tracking data, can provide insights into learners' visual attention and cognitive load during problem-solving tasks in engineering games (Tsai et al. 2019). By analyzing gaze patterns, the game can

adapt its content presentation or difficulty level in real time, ensuring that learners are appropriately challenged without becoming overwhelmed. Similarly, electroencephalography (EEG) data can assess cognitive workload and emotional states, allowing for more nuanced adaptations in-game mechanics or instructional support (Ahonen et al. 2021).

Behavioral data, such as mouse movements, keystrokes, and in-game actions, can also be utilized to create more personalized learning experiences. Shute & Rahimi (2022) demonstrated that analyzing patterns in learners' problem-solving approaches in physics-based games could provide valuable insights into their conceptual understanding and skill development. By leveraging this data, GBLEs can offer personalized feedback, hints, or additional practice opportunities tailored to each learner's needs and progress in engineering concepts.

Affective computing techniques can enhance adaptivity by recognizing and responding to learners' emotional states. Facial expression analysis and speech recognition can detect frustration, engagement, or confusion during gameplay (D'Mello and Graesser 2015; Giannakos et al. 2022). This emotional data can be combined with performance metrics to understand the learner's state better, allowing the game to adapt its pace, difficulty, or feedback style. For example, if a learner shows frustration while attempting to solve a complex engineering problem, the game could offer more scaffolding or break the problem down into smaller, more manageable steps.

Multimodal data plays a crucial role in creating more accurate *learner models*, which is essential for effective personalization and adaptivity. ML algorithms can create more comprehensive and dynamic learner profiles by combining data from various sources, such as performance metrics, physiological measures, and behavioral patterns (Chrysafiadi and Virvou 2013). These enhanced learner models can then be used to make more informed decisions about content selection, difficulty adjustment, and instructional strategies within the GBLE, providing reassurance about the effectiveness of the adaptive mechanisms.

Furthermore, there is great potential for multimodal data to support collaborative learning experiences in engineering education. By analyzing

speech patterns, facial expressions, and collaborative problem-solving behaviors, GBLEs can adapt to group dynamics and provide targeted support for effective teamwork (Spikol et al. 2018). This is particularly relevant in engineering education, where collaborative skills are essential for success in professional practice, and the use of multimodal data can significantly enhance the learning experience.

Integrating multimodal data can also enhance the assessment capabilities of GBLEs in engineering education. Stealth assessment techniques, which unobtrusively collect data on learner performance and behavior throughout the game, can be augmented with multimodal data to provide more accurate and comprehensive evaluations of learners' skills and knowledge (Shute and Rahimi 2022). This approach allows for continuous, formative assessments that can inform the game's adaptive mechanisms and provide valuable feedback to instructors and learners.

However, it is essential to note that using multimodal multichannel data in GBLEs also presents challenges. These include technical issues related to data integration and real-time processing and ethical considerations regarding data privacy and consent (Blikstein and Worsley 2016). Researchers and developers must carefully consider these factors when designing and implementing multimodal adaptive systems for engineering education.

In conclusion, the integration of multimodal multichannel data holds immense potential to significantly enhance personalization and adaptivity in GBLEs for engineering education. By providing a more comprehensive understanding of learners' cognitive, affective, and behavioral states, this approach can lead to more effective, engaging, and tailored learning experiences. As technology advances, the potential for leveraging multimodal data in educational games will likely expand, offering new opportunities to improve engineering education through adaptive and personalized learning experiences.

11.6 AI Techniques for Handling Multimodal Approaches to Individualization and Adaptation

AI has revolutionized personalized learning (du Boulay et al. 2023) and engineering education by offering innovative solutions that cater to individual learning needs and preferences. Integrating AI in educational settings enhances the learning experience by providing tailored content, real-time feedback, and adaptive learning paths. This section describes multimodal approaches to individualization and adaptation in personalized learning and engineering education, focusing on how AI leverages multimodal data sources and methodologies to create effective and engaging GBLEs.

AI technologies can enable personalized learning by analyzing vast amounts of data to understand each learner's strengths, weaknesses, preferences, and progress. This data-driven approach allows AI systems to adapt instructional content and strategies to meet each student's unique needs. Essential AI techniques in personalized learning include ML, natural language processing (NLP), generative AI (gen-AI), and computer vision. In this section, we briefly describe how each AI technique addresses the issues of personalization and adaptivity (see Lester et al. 2024).

ML algorithms analyze data from learners' interactions with educational content to identify patterns and predict their learning needs. For example, adaptive learning platforms, which are AI-driven systems that adjust the learning experience based on a student's progress and performance, use ML to recommend resources most likely to help students overcome specific challenges. AI systems ensure learners receive the most relevant and practical support by continuously updating these recommendations based on real-time performance data.

NLP enables AI systems to understand and generate human language, making it possible to create intelligent tutoring systems and conversational agents that provide personalized feedback and support. For instance, AI-driven pedagogical agents and nonplayer characters can answer students' questions, guide them through complex problems, and offer expla-

nations conversationally, making learning more interactive and accessible.

Gen-AI refers to how AI can be used to create new content, such as text, images, music, etc., based on data and information from other sources, leveraging and converging the power of generative adversarial networks and large language models. Gen-AI makes it possible to embed novel content that is personalized to the individual learner. For example, GBLEs empowered by Gen-AI can generate practice quizzes, explanations from pedagogical agents, and new scenarios that support existing curricula but are adapted toward students' growing knowledge from the GBLE.

Computer vision technology allows AI systems to analyze visual data, such as facial expressions and gestures, to gauge learners' emotional states and engagement levels. This information can adjust instructional strategies in real time, providing additional support or challenges as needed to maintain optimal engagement and motivation.

Multimodal approaches leverage multiple data sources and modalities to understand each learner comprehensively. These approaches integrate information from various channels, such as textual data, audio–visual inputs, and physiological signals, to deliver highly personalized learning experiences.

Textual data analysis, including students' written responses, forum posts, and digital notes, provides insights into their cognitive processes and comprehension levels. AI systems can use this information to tailor feedback and suggest additional reading materials or exercises that target specific areas of improvement. *Audio–visual inputs offer* valuable data on learners' engagement, participation, and understanding. For example, AI can analyze video recordings of classroom activities to identify students needing additional support or to recognize patterns that indicate common areas of difficulty. By incorporating audio–visual data, AI systems can create more dynamic and responsive GBLEs.

Physiological signals captured by wearable devices and sensors (e.g. electrodermal activity bracelets and EEG headsets) provide real-time data on learners' emotional states and attention levels. AI systems can use this

data to detect frustration, boredom, or anxiety and adjust instructional strategies to enhance learning outcomes. For example, if a student exhibits signs of stress, the system might offer a more straightforward task or a short break to help them regain focus.

Integrating AI in GBLEs and engineering education through multimodal approaches to individualization and adaptation offers significant potential for enhancing learning outcomes and experiences. By leveraging ML, NLP, computer vision, and physiological signal analysis, AI-based GBLEs can create highly responsive and tailored educational environments. These innovations address the diverse needs of learners and support the development of SRL skills such as goal setting, time management, and self-assessment, and engineering knowledge and skills required for success in engineering and other technical fields.

11.7 Essential SRL Processes from Multimodal Trace Data with GBLEs in Engineering Education

The use of SRL strategies during game-based learning are essential for increased learning outcomes; therefore, it is critical for GBLEs to identify students' abilities to engage in SRL during game-based learning, track this over time, and personalize and adapt the learning experience to meet individual learner needs. Among the most essential and effective SRL strategies are goal setting, self-monitoring, strategic planning, self-reflection, time management, help-seeking, adaptivity, personalization features, and resource management (Azevedo and Dever 2022; Winne and Hadwin 1998; Winne 2018; Winne and Azevedo 2022; Zimmerman and Moylan 2009). This section introduces the eight SRL processes essential to learning, reasoning, and problem-solving with GBLEs across engineering education and describes how each one can be detected, modeled, and traced during game-based learning in order to personalize and adapt to each learner's needs. We acknowledge that there are other SRL processes, but it is impossible to discuss them all due to space limitations (e.g. see Azevedo et al. 2024).

11.7.1 Goal Setting

Goal setting is not just a component but a cornerstone of effective learning, particularly in GBLEs for engineering education. It is a topic that should engage and interest all those involved in game-based learning. Detecting, modeling, tracing, and using goal-setting behaviors to adapt and personalize these environments can significantly enhance the learning experience (Locke and Latham 2002; Pintrich 2000). By leveraging multimodal trace data, including log files, think-aloud protocols, eye movements, facial expressions, learner-system recordings, and physiological signals, researchers and educators can gain valuable insights into students' goal-setting processes and tailor the learning experience accordingly (D'Mello et al. 2017).

Detecting goal-setting behaviors involves analyzing various data sources. Log files can track user interactions, such as menu selections and navigation patterns, revealing learners' priorities and objectives (Kostons et al. 2012). Think-aloud protocols provide direct verbal expressions of intentions and plans, offering a window into students' thought processes (Ericsson and Simon 1993). Eye-tracking data can show where learners focus their attention, potentially indicating goal-related interests (Halszka et al. 2017). Facial expressions and physiological signals can offer insights into emotional states and arousal levels associated with goal-setting and pursuit (Azevedo et al. 2018). Learner-system recordings capture actions and decisions within the game environment, reflecting goal-oriented behaviors (Shute and Ventura 2017).

Once detected, goal-setting behaviors can be modeled using various techniques. ML algorithms can classify goals, such as performance or learning goals (Conati and Karden 2013). Sequence models like recurrent neural networks (RNNs) can capture the temporal aspects of goal-setting and pursuit (Molenaar and Knoop-van Campen 2019). Bayesian networks can represent the probabilistic relationships between goals, actions, and outcomes, providing a comprehensive model of the goal-setting process (Conati et al. 2018).

Tracing goal-setting involves continuous monitoring and analysis of the ongoing process. Real-time analysis of multimodal data allows for dynam-

ic updates to goal models (Blikstein and Worskley 2018). Visualization tools can give learners and instructors graphical representations of goal progress, enhancing self-awareness and facilitating intervention (Bull and Kay 2016). Intelligent tutoring systems can integrate goal tracing into adaptive support mechanisms, offering personalized guidance based on the learner's current goal state and performance (VanLehn 2011).

The insights gained from detecting, modeling, and tracing goal-setting can be used to adapt and personalize the game-based learning experience. AI plays a crucial role in this process, ensuring not only that each learner feels valued and catered to based on various levels of self-efficacy or goal-orientations (i.e. the affective and motivational processes in SRL), but also that the learner is able to evaluate and monitor their overall learning progression in relationships to their larger goals outside of the immediate learning session (i.e. the (meta)cognitive processes in SRL).

Adaptive and personalized support within GBLEs centered on goal-setting take multiple forms. Dynamic difficulty adjustment can modify challenge levels based on the learner's goals and capabilities (Sampayo-Vargas et al. 2013). Personalized feedback can be tailored to align with individual goal orientations, providing more process-oriented or outcome-oriented guidance (Hattie and Timperley 2007). Adaptive content sequencing can organize learning materials and activities to support goal achievement, while motivational interventions can help maintain engagement and goal commitment (Kickmeier-Rust and Albert 2010).

11.7.2 Self-monitoring

Self-monitoring, another critical metacognitive process in learning, holds particular significance in GBLEs in engineering education. It entails learners actively observing and evaluating their own cognitive processes, progress, and performance (Winne 2018; Zimmerman and Moylan 2009). By harnessing multimodal trace data, researchers and educators can gain valuable insights into students' self-monitoring behaviors and use this information to adapt and personalize the learning experience (Dever et al. in press).

The detection of self-monitoring behaviors can be achieved through various data sources. Log files can unveil patterns of self-checking activities, such as reviewing completed tasks or accessing performance metrics (Winne and Hadwin 2013). Think-aloud protocols provide direct verbal evidence of self-monitoring, capturing learners' reflections on their understanding and progress (Azevedo et al. 2004). Eye-tracking data can indicate moments of self-evaluation, such as fixations on feedback or progress indicators (Dever et al. 2023). Facial expressions can reveal emotional responses to self-assessment, while physiological signals like heart rate variability or skin conductance can indicate cognitive load during monitoring activities (D'Mello et al. 2017). The role of AI in processing and interpreting these data sources is crucial in this context, as it can handle the complexity and volume of data, identify patterns, and make predictions, thereby enhancing the understanding of self-monitoring behaviors.

Modeling self-monitoring behaviors involves creating computational representations of these processes. ML algorithms can classify self-monitoring activities based on multimodal data (Azevedo et al. 2018). For example, a support vector machine (SVM) might distinguish between surface-level monitoring (e.g. checking task completion without considering the quality of the work) and deep monitoring (e.g. evaluating understanding and the quality of the work). Temporal models, such as hidden Markov models, can capture the sequential nature of self-monitoring processes, revealing how learners transition between different monitoring states (Dever et al. 2023a; Molenaar and Järvelä 2014).

Tracing self-monitoring involves real-time tracking and analysis of these behaviors as they occur during the learning process. This can be achieved by continuously processing multimodal data streams and updating learner models on the fly (Blikstein and Worsley 2016). Visualization tools can provide learners and instructors with real-time representations of self-monitoring activities, promoting awareness and facilitating timely interventions (Wiedbusch et al. 2021).

The insights gained from detecting, modeling, and tracing self-monitoring can be used to adapt and personalize the game-based learning experience. Adaptive scaffolding can be implemented to support learners who

demonstrate inadequate self-monitoring, such as providing prompts for reflection or offering structured self-assessment tools (Azevedo et al. 2016). Conversely, the system might reduce explicit support and increase autonomy for learners exhibiting effective self-monitoring. The game environment can be dynamically adjusted to match the learner's current monitoring state, for instance, by modifying the frequency and detail of feedback or adjusting the complexity of tasks based on the learner's demonstrated ability to self-evaluate (Taub et al. 2014).

11.7.3 Strategic Planning

Strategic planning is a critical component of SRL, particularly in GBLEs for engineering education. It involves selecting appropriate strategies, and allocating resources to achieve desired outcomes based on set goals (Zimmerman and Moylan 2009). By leveraging multimodal trace data, researchers and educators can gain valuable insights into students' strategic planning processes and use this information to adapt and personalize the learning experience.

Detecting strategic planning behaviors can be accomplished through various data sources. Log files can reveal patterns of resource allocation, task sequencing, and tool selection that indicate deliberate planning (Winne and Hadwin 2013). Think-aloud protocols provide direct verbal evidence of planning, capturing learners' thoughts on strategy selection and goal setting (Greene and Azevedo 2009). Eye-tracking data can indicate moments of strategic consideration, such as scanning available resources or reviewing task requirements before taking action (Wiedbusch et al. 2020). Facial expressions can reveal emotional responses during the planning process, while physiological signals like pupil dilation or skin conductance can indicate cognitive engagement in planning activities (Harley et al. 2015).

Modeling strategic planning behaviors involves creating computational representations of these processes. ML algorithms, such as decision trees or neural networks, can be employed to classify different types of planning strategies based on multimodal data (Azevedo et al. 2018). For example, a random forest (RF) classifier might distinguish between surface-lev-

el planning (e.g. simple task sequencing) and deep planning (e.g. considering multiple solution paths). Temporal models, such as RNNs, can capture the sequential nature of planning processes, revealing how learners develop and adjust their strategies over time.

Tracing strategic planning involves real-time tracking and analysis of these behaviors as they occur during the learning process. This can be achieved through continuous processing of multimodal data streams, updating learner models on-the-fly (Blikstein and Worsley 2016). Visualization tools can provide both learners and instructors with real-time representations of planning activities, promoting awareness and facilitating timely interventions (Wiedbusch et al. 2024).

The insights gained from detecting, modeling, and tracing strategic planning can be used to adapt and personalize the game-based learning experience. Adaptive scaffolding can be implemented to support learners who demonstrate inadequate planning skills, such as providing structured planning tools or suggesting effective strategies (Azevedo et al. 2016). Conversely, for learners exhibiting sophisticated planning abilities, the system might reduce explicit support and increase the complexity of planning challenges. The game environment can be dynamically adjusted to match the learner's current planning state, for instance, by modifying the availability of resources or adjusting the time pressure based on the learner's demonstrated ability to plan effectively (Taub et al. 2014).

11.7.4 Self-reflection

Self-reflection is an essential metacognitive process in learning, particularly relevant in GBLEs for engineering education. It involves learners actively examining their own thoughts, experiences, and performance to gain insights and improve future learning (Zimmerman 2002). By leveraging multimodal trace data, researchers and educators can gain valuable insights into students' self-reflective behaviors and use this information to adapt and personalize the learning experience.

Detecting self-reflective behaviors can be accomplished through various data sources. Log files can reveal patterns of revisiting completed tasks,

reviewing feedback, or accessing reflective prompts, indicating moments of self-reflection (Winne and Hadwin 2013). Think-aloud protocols provide direct verbal evidence of self-reflection, capturing learners' thoughts on their performance, strategies, and areas for improvement (Azevedo and Cromley 2004). Eye-tracking data can indicate reflective moments, such as extended fixations on performance summaries or areas of difficulty (Wiedbusch et al. 2023). Facial expressions can reveal emotional responses during self-reflection, while physiological signals like heart rate variability or skin conductance can indicate cognitive engagement in reflective activities (Harley et al. 2019).

Modeling self-reflective behaviors involves creating computational representations of these processes. ML algorithms, such as SVMs or deep neural networks, can be employed to classify different types of reflective activities based on multimodal data (Azevedo et al. 2018). For example, a classifier might distinguish between surface-level reflection (e.g. simple performance review) and deep reflection (e.g. analyzing causal relationships between actions and outcomes). NLP techniques can be applied to analyze the content and depth of reflective statements in think-aloud protocols or written reflections (Kovanović et al. 2023).

Tracing self-reflection involves real-time tracking and analysis of these behaviors as they occur during the learning process. This can be achieved through continuous processing of multimodal data streams, updating learner models on-the-fly (Blikstein and Worsley 2016). Visualization tools can provide both learners and instructors with real-time representations of reflective activities, promoting awareness and facilitating timely interventions (Bull and Kay 2016).

The insights gained from detecting, modeling, and tracing self-reflection can be used to adapt and personalize the game-based learning experience. Adaptive scaffolding can be implemented to support learners who demonstrate inadequate reflection skills, such as providing structured reflective prompts or guiding questions (Lin et al. 1999). Conversely, for learners exhibiting sophisticated reflective abilities, the system might reduce explicit support and increase the complexity of reflective challenges. The game environment can be dynamically adjusted to match the learn-

er's current reflective state, for instance, by modifying the frequency and depth of reflective opportunities or adjusting the level of metacognitive support based on the learner's demonstrated ability to reflect effectively (Taub et al. 2014).

In the context of engineering education, self-reflection can be particularly valuable for developing problem-solving skills and design thinking. AI systems can be designed to prompt reflection on specific engineering concepts, design decisions, or problem-solving strategies, helping students connect theoretical knowledge with practical applications (Turns et al. 2014). For example, after completing a virtual design task, the system might guide students through a reflective process that considers not only the technical aspects of their solution but also its potential social and environmental impacts, aligning with the broader goals of engineering education (Baillie and Armstrong 2013).

11.7.5 Time Management

Time management is a critical skill in engineering education and professional practice, particularly relevant in GBLEs where students must balance multiple tasks and objectives. Effective time management involves planning, prioritizing, and allocating time to various activities to maximize productivity and learning outcomes. This is supported by studies such as Claessens et al. (2007), which found that effective time management is a key factor in academic success. By leveraging multimodal trace data, including log files, think-aloud protocols, eye movements, facial expressions, learner-system recordings, and physiological signals, as demonstrated in various studies (e.g. Azevedo et al. 2011; Harley et al. 2021), researchers and educators can gain valuable insights into students' time management behaviors and use this information to adapt and personalize the learning experience.

Understanding time management behaviors can be achieved by capturing various data sources. Log files can reveal patterns of task engagement, task switching, and time spent on different activities, indicating how learners allocate their time (Winne 2018). Think-aloud protocols provide direct verbal evidence of time management strategies, capturing

learners' thoughts on prioritization and time allocation (Azevedo et al. 2011). Eye-tracking data can indicate time-related behaviors, such as frequent glances at a clock or timer, or scanning of task lists (Azevedo et al. 2022). Facial expressions can reveal emotional responses to time pressure or successful time management, while physiological signals like heart rate variability or skin conductance can indicate stress levels related to time constraints (Harley et al. 2021).

Modeling time management behaviors involves creating computational representations of these processes. ML algorithms, such as decision trees or neural networks, can be employed to classify different time management strategies based on multimodal data (Azevedo et al. 2018). For example, a classifier might distinguish between proactive time management (e.g. creating schedules and setting deadlines) and reactive time management (e.g. rushing to complete tasks as deadlines approach). Temporal models, such as hidden Markov models or RNNs, can capture the sequential nature of time management processes, revealing how learners adjust their strategies over time (Järvelä and Bannert 2021).

Tracing time management involves real-time tracking and analysis of these behaviors as they occur during the learning process. This can be achieved through continuous processing of multimodal data streams, updating learner models on-the-fly (Giannakos et al. 2022). Visualization tools can provide both learners and instructors with real-time representations of time management activities, promoting awareness and facilitating timely interventions (Molenaar 2021).

The insights gained from detecting, modeling, and tracing time management can be used to adapt and personalize the GBL experience. This personalization is a key value, as it allows for adaptive scaffolding to support learners who demonstrate poor time management skills, such as providing time management tips or visual cues for task durations (Taub et al. 2014). Conversely, for learners exhibiting effective time management, the system might introduce more complex time-constrained challenges. The game environment can be dynamically adjusted to match the learner's current time management state, for instance, by modifying task deadlines

or introducing unexpected events that require rapid reprioritization (Kickmeier-Rust and Albert 2010).

In the context of engineering education, effective time management is crucial for project-based learning and complex problem-solving tasks. AI systems can be designed to simulate real-world engineering scenarios with time constraints, helping students develop time management skills that are directly applicable to their future careers (Jonassen et al. 2006). For example, a GBLE might present students with a multiphase engineering design project, requiring them to allocate time effectively across research, design, prototyping, and testing phases while managing unexpected challenges and resource constraints. This practical application of AI in education is a key aspect of its relevance and potential impact.

11.7.6 Help-seeking

Help-seeking is a crucial aspect of SRL, particularly relevant to GBLEs for engineering education. It involves recognizing when assistance is needed, identifying potential sources of help, and effectively utilizing available resources (Karabenick and Berger 2013). By leveraging multimodal trace data, including log files, think-aloud protocols, eye movements, facial expressions, learner-system recordings, and physiological signals, researchers and educators can gain valuable insights into students' help-seeking behaviors and use this information to adapt and personalize the learning experience.

Detecting help-seeking behaviors can be accomplished through various data sources. Log files can reveal patterns of accessing help features, consulting resources, or seeking peer assistance within the game environment (Winne and Hadwin 2011). Think-aloud protocols provide direct verbal evidence of help-seeking intentions and strategies, capturing learners' thoughts on when and how to seek assistance (Azevedo et al. 2014). Eye-tracking data can indicate help-seeking moments, such as scanning for help buttons or focusing on areas of difficulty (Taub et al. 2019). Facial expressions can reveal emotional states associated with confusion or frustration that may precede help-seeking. At the same time, physiological signals like increased heart rate or skin conductance can indicate

stress or cognitive load that might trigger help-seeking behavior (Harley et al. 2017).

Modeling help-seeking behaviors involves creating computational representations of these processes. ML algorithms, such as decision trees or SVMs, can classify different help-seeking strategies based on multimodal data (Azevedo et al. 2019). For example, a classifier might distinguish between instrumental help-seeking (seeking hints or guidance) and executive help-seeking (seeking direct answers or solutions). Temporal models, such as hidden Markov models or RNNs, can capture the sequential nature of help-seeking processes, revealing how learners' strategies evolve (Molenaar and Järvelä 2014).

Tracing help-seeking involves real-time tracking and analysis of these behaviors during the learning process. This can be achieved by continuously processing multimodal data streams and updating learner models on the fly (Blikstein and Worsley 2016). Visualization tools can provide both learners and instructors with real-time representations of help-seeking activities, promoting awareness and facilitating timely interventions (Bull and Kay 2016).

The insights gained from detecting, modeling, and tracing help-seeking can be used to adapt and personalize the GBL experience. Adaptive scaffolding can be implemented to support learners who demonstrate ineffective help-seeking behaviors, such as providing explicit cues for available help or guidance on formulating effective help requests (Aleven et al. 2016). Conversely, for learners exhibiting sophisticated help-seeking strategies, the system might introduce more challenging scenarios requiring more nuanced assistance. The game environment can be dynamically adjusted to match the learner's current help-seeking state, for instance, by modifying the availability or format of help resources based on the learner's demonstrated ability to seek and utilize assistance effectively (Koedinger et al. 2013).

In the context of engineering education, effective help-seeking is crucial for problem-solving and collaborative work. AI systems can be designed to simulate real-world engineering scenarios where seeking expertise from colleagues or consulting technical documentation is essential

(Jonassen et al. 2006). For example, a GBL environment might present students with a complex engineering design task that requires them to identify when they need assistance, choose appropriate sources of help (e.g. virtual mentors, technical manuals, or peer collaboration), and effectively apply the received guidance to solve the problem.

11.7.7 Resource Management

Effective resource management involves planning, prioritizing, and optimizing available materials, tools, time, and information to achieve desired outcomes (Azevedo and Hadwin 2005). Resource management is a critical skill in engineering education and professional practice, particularly relevant in GBLEs, where students must efficiently allocate and utilize various resources to solve complex problems. By leveraging multimodal trace data, including log files, think-aloud protocols, eye movements, facial expressions, learner-system recordings, and physiological signals, researchers and educators can gain valuable insights into students' resource management behaviors and use this information to adapt and personalize the learning experience.

Detecting resource management behaviors can be accomplished through various data sources. Log files can reveal resource access, utilization, and allocation patterns within the game environment (Biswas et al. 2016). Think-aloud protocols provide direct verbal evidence of resource management strategies, capturing learners' thoughts on prioritization and decision-making regarding resource use (Greene et al. 2018). Eye-tracking data can indicate resource-related behaviors, such as scanning available resources, comparing options, or focusing on resource-constrained areas of a problem (Wiedbusch et al. 2024). Facial expressions can reveal emotional responses to resource constraints or successful resource allocation. At the same time, physiological signals like heart rate variability or skin conductance can indicate stress levels related to resource management challenges (Taub et al. 2021).

Modeling resource management behaviors involves creating computational representations of these processes. ML algorithms, such as RF or neural networks, can classify different resource management strategies

based on multimodal data (Azevedo et al. 2018). For example, a classifier might distinguish between proactive resource planning (e.g. strategic allocation of resources at the beginning of a task) and reactive resource management (e.g. adjusting resource allocation in response to emerging challenges). Temporal models, such as RNNs or dynamic Bayesian networks, can capture the sequential nature of resource management processes, revealing how learners adjust their strategies over time (Nguyen et al. 2023).

Tracing resource management involves real-time tracking and analysis of these behaviors during learning. This can be achieved by continuously processing multimodal data streams and updating learner models on the fly (Blikstein and Worsley 2016). Visualization tools can provide learners and instructors with real-time representations of resource management activities, promoting awareness and facilitating timely interventions (Bull and Kay 2016).

The insights gained from detecting, modeling, and tracing resource management can be used to adapt and personalize the game-based learning experience. Adaptive scaffolding can be implemented to support learners who demonstrate poor resource management skills, such as providing resource management tips or visual cues for optimal resource allocation (Taub et al. 2014). Conversely, the system might introduce more complex resource-constrained challenges for learners exhibiting effective resource management. The game environment can be dynamically adjusted to match the learner's current resource management state, for instance, by modifying resource availability or introducing unexpected events that require rapid resource reallocation (Kickmeier-Rust and Albert 2010).

Effective resource management is crucial for project-based learning and complex problem-solving tasks in engineering education. AI systems can be designed to simulate real-world engineering scenarios with resource constraints, helping students develop resource management skills directly applicable to their future careers (Jonassen et al. 2006). For example, a GBLE might present students with a multiphase engineering design project, requiring them to manage limited materials, budget, time, and

human resources while optimizing for performance, cost, and sustainability.

In summary, addressing all these SRL challenges will require interdisciplinary collaboration among learning scientists, engineering educators, game designers, AI researchers, and engineering professionals. As AI technology advances and our understanding of the dynamic nature of SRL processes during gameplay with GBLEs deepens, the potential for AI to enhance SRL skills in educational games continues to grow, promising more effective and personalized learning experiences for engineering students.

11.8 Open Questions, Future Directions, and Conclusions

The use of AI and multimodal multichannel data in GBLEs for engineering education not only presents numerous opportunities for future research but also holds the potential to significantly advance the field. Particularly, in addressing adaptivity and personalization, this research could pave the way for transformative changes. Several major questions emerge as critical areas for investigation, each with the potential to shape the future of engineering education. Below, we list a few of these questions and highlight their importance.

How can we effectively integrate and analyze multimodal data from GBLEs to create more accurate and comprehensive learner models? This question, with its inherent complexity, addresses the challenge of combining diverse data sources such as gameplay interactions, physiological responses, and verbal or written inputs to form a holistic understanding of the learner's cognitive and affective states and as a self-regulating agent. Future research should explore advanced ML techniques, including deep learning and multitask learning, to process and interpret this complex data effectively, engaging researchers in a challenging yet rewarding pursuit.

What AI algorithms and architectures are most effective for real-time adaptation in engineering education GBLEs? This question focuses on develop-

ing AI systems that can make rapid, context-appropriate adjustments to the learning experience. Researchers should investigate reinforcement learning approaches, dynamic difficulty adjustment algorithms, and adaptive pedagogical agents to create more responsive and personalized learning environments.

How can we design AI-driven personalization systems that effectively support the development of complex engineering skills and problem-solving abilities? This question addresses the unique challenges of engineering education, where learners must master abstract concepts and apply them to real-world problems. Future research should explore knowledge tracing algorithms specific to engineering domains and investigate how AI can support the scaffolding of complex problem-solving processes.

How can we design AI systems that promote transfer of learning and long-term retention of engineering concepts in GBLEs? This question addresses the ultimate goal of educational interventions – to facilitate learning that extends beyond the immediate context. Researchers should investigate how AI can be used to identify opportunities for knowledge transfer, provide spaced repetition of key concepts, and support metacognitive skill development.

How can we develop scalable and generalizable AI models for adaptive learning that can be effectively applied across different engineering disciplines and educational contexts? This question addresses the need for flexible and robust AI-based GBLEs that can be deployed in various settings. Future research should explore transfer learning techniques, domain adaptation methods, and modular AI architectures that can be easily customized for different engineering fields and learning objectives.

How can we ensure the ethical implementation of AI-driven adaptive systems in engineering education, particularly regarding fairness, transparency, and privacy? This question, of utmost importance, addresses the critical need for responsible AI development in educational contexts. Future studies should focus on developing frameworks for ethical AI in education, methods for detecting and mitigating bias in adaptive systems, and approaches to balance personalization with privacy protection. It is

our collective responsibility to ensure that AI is used in a way that respects the rights and dignity of learners.

What AI algorithms and architectures are most effective for adapting GBLEs to support individual differences in SRL among engineering students? This question underscores the urgency of developing AI systems that can identify and respond to varying levels of SRL skills and strategies. It is a significant challenge given the quantitative (e.g. increased number of learning strategies, duration of each metacognitive process) and qualitative (e.g. accurate metacognitive monitoring) nature of SRL skills and strategies. Researchers should investigate reinforcement learning approaches and adaptive pedagogical agents that can provide personalized scaffolding for different phases of SRL, such as goal setting, monitoring, and reflection.

How can we design AI-driven systems that effectively balance explicit instruction and implicit support for SRL in engineering GBLEs? This question emphasizes the need for continuous research to explore how AI can dynamically adjust the level and type of SRL support based on learners' progress and needs. It addresses the need to develop learners' SRL skills while maintaining engagement with the engineering content, especially when dealing with challenging and complex topics.

How can AI effectively leverage multimodal data, such as gameplay interactions, physiological responses, verbal protocols, eye movements, and facial expressions, to accurately model and support SRL processes in engineering GBLEs? This question addresses the challenge of integrating diverse data sources to understand and scaffold learners' metacognitive processes. Further complicating this issue is the lack of prescriptive methods needed to determine the timing, type, and sequence of scaffolding methods and techniques and to determine their impact, effectiveness, and efficacy to each individual learner. Future research should explore advanced ML techniques, such as deep learning and multitask learning, to process this complex data and provide real-time support for SRL strategies.

In conclusion, this chapter has examined the transformative role of AI in personalizing and adapting engineering education through GBLEs. We have explored multimodal approaches that leverage AI to create tailored,

interactive educational experiences, addressing individual learners' educational and SRL competencies, needs and challenges. Key features of personalization and adaptivity were discussed, supported by the collection of multimodal, multichannel SRL trace data, such as log files, eye movements, facial expressions of emotions, physiological data, concurrent verbalizations, and audio and video screen recordings of learner-GBLE interactions. These data drive personalization and adaptivity, aiming to maximize learning, problem-solving, and the understanding of complex engineering concepts and phenomena. We also presented, defined, and provided examples of seven specific SRL processes and how they can be detected, modeled, and traced using various SRL trace methodologies to understand the SRL challenges experienced by individual learners. Lastly, we highlighted several open questions, future directions, and conclusions, offering practical recommendations for implementing AI-enhanced game-based learning strategies.

Acknowledgments

This chapter was partially supported by funding from the National Science Foundation (DRL #1916417, BCS #2128684, IIS #2302778) and the National Institutes of Health (1R25GM150142-01). Any opinions, findings, conclusions, or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation or the National Institutes of Health.

References

Ahonen, V., Leino, M., and Lipping, T. (2021). Electroencephalography in evaluating mental workload of gaming. In: *Annual International Conference of the IEEE Engineering Medical and Biological Society* (pp. 845–848). https://doi.org/10.1109/EMBC46164.2021.9629772.

Aleven, V., Roll, I., McLaren, B.M., et al. (2016). Help helps, but only so much: research on help seeking with intelligent tutoring systems. *International Journal of Artificial Intelligence in Education* 26 (1): 205–223.

Azevedo, R. (2015). Defining and measuring engagement and learning in science: Conceptual, theoretical, methodological, and analytical issues.

Educational Psychologist 50: 84–94.

Azevedo, R. and Cromley, J.G. (2004). Does training on self-regulated learning facilitate students' learning with hypermedia? *Journal of Educational Psychology* 96: 523–535.

Azevedo, R. and Dever, D. (2022). Metacognition in multimedia learning. In: *Cambridge handbook of multimedia*, (ed. R.E. Mayer and L. Fiorella) (3rd ed., pp. 132–141). Cambridge University Press.

Azevedo, R. and Gasevic, D. (2019). Analyzing multimodal multichannel data about self-regulated learning with advanced learning technologies: issues and challenges. *Computers in Human Behavior* 96: 207–210.

Azevedo, R. and Hadwin, A.F. (2005). Scaffolding self-regulated learning and metacognition – implications for the design of computer-based scaffolds. *Instructional Science* 33 (5-6): 367–379.

Azevedo, R. and Wiedbusch, M. (2023). Theories of metacognition and pedagogy applied in AIED systems. In: *Handbook of Artificial Intelligence in Education* (ed. B. du Boulay, T. Mitrovic, and K. Yacef) (pp. 141–173). Edward Elgar Publishing.

Azevedo, R., Amon, M.J., Anderson, M. et al. (2024). Human digital twins to support nurse practitioners' clinical decision-making using multimodal data: A theoretical, methodological, and analytical framework. In: *Digital Twin: Fundamentals and Applications* (ed. S. Sabri, K. Alexandridis, and N. Lee) (pp. 149–172). Springer/Nature.

Azevedo, R., Bouchet, F., Harley, J. et al. (2022). Lessons learned and future directions of MetaTutor: Leveraging multichannel data to scaffold self-regulated learning with an intelligent tutoring system. *Frontiers in Psychology*: 13: 813632. https://doi.org/10.3389/fpsyg.2022.813632.

Azevedo, R., Cromley, J.G., and Seibert, D. (2004). Does adaptive scaffolding facilitate students' ability to regulate their learning with hypermedia? *Contemporary Educational Psychology* 29 (3): 344–370.

Azevedo, R., Cromley, J.G., Moos, D.C. et al. (2011). Adaptive content and process scaffolding: A key to facilitating students' self-regulated learning with hypermedia. *Psychological Testing and Assessment Modeling* 53: 106–140.

Azevedo, R., Harley, J., Trevors, G., et al. (2013). Using trace data to examine the complex roles of cognitive, metacognitive, and emotional self-regulatory processes during learning with multi-agent systems. In: *International Handbook of Metacognition and Learning Technologies*,

(ed. R. Azevedo and V. Aleven), (Vol. 28). Springer International Handbooks of Education. Springer. https://doi.org/10.1007/978-1-4419-5546-3_28.

Azevedo, R., Mudrick, N.V., Taub, M., et al. (2019). Self-regulation in computer-assisted learning systems. In: *Handbook of Cognition and Education*, (ed. J. Dunlosky and K. Rawson), (pp. 587–618). Cambridge University Press.

Azevedo, R., Taub, M., and Mudrick, N.V. (2018). Understanding and reasoning about real-time cognitive, affective, and metacognitive processes to foster self-regulation with advanced learning technologies. In: *Handbook of Self-regulation of Learning and Performance*, 2e, (ed. D.H. Schunk and J.A. Greene) (pp. 254–270). Routledge.

Baillie, C. and Armstrong, R. (2013). Crossing knowledge boundaries and thresholds: challenging the dominant discourse within engineering education. In: *Engineering Education for Social Justice*, (ed. J. Lucena), (pp. 135–152). Springer.

Bernold, L.E., Spurlin, J.E., and Anson, C.M. (2007). Understanding our students: A longitudinal study of success and failure in engineering with implications for increased retention. *Journal of Engineering Education* 96 (3): 263–274. https://doi.org/10.1002/j.2168-9830.2007.tb00935.x.

Biswas, G., Segedy, J.R., and Bunchongchit, K. (2016). From design to implementation to practice: A learning by teaching system: Betty's Brain. *International Journal of Artificial Intelligence in Education* 26 (1): 350–364. https://doi.org/10.1007/s40593-015-0057-9.

Blikstein, P. and Worsley, M. (2016). Multimodal learning analytics and education data mining: using computational technologies to measure complex learning tasks. *Journal of Learning Analytics* 3 (2): 220–238.

Bull, S. and Kay, J. (2016). SMILI ☺: a framework for interfaces to learning data in open learner models, learning analytics and related fields. *International Journal of Artificial Intelligence in Education* 26 (1): 293–331.

Butler, S. and Ahmed, D. (2016). *Gamification to Engage and Motivate Students to Achieve Computer Science Learning Goals* (pp. 237–240). https://doi.org/10.1109/CSCI.2016.0053.

Chen, X., Zou, D., Cheng, G., et al. (2021). Detecting latent topics and trends in educational technologies over four decades using structural

topic modeling: a retrospective of all volumes of Computers & Education. *Computers & Education* 169: 104222.

Chrysafiadi, K. and Virvou, M. (2013). Student modeling approaches: a literature review for the last decade. *Expert Systems with Applications* 40 (11): 4715–4729.

Claessens, B.J., Van Eerde, W., Rutte, C.G., et al. (2007). A review of the time management literature. *Personnel Review* 36 (2): 255–276.

Clark, D.B., Tanner-Smith, E.E., Killingsworth, S.S. (2016). Digital games, design, and learning: a systematic review and meta-analysis. *Review of Educational Research* 86 (1): 79–122.

Coller, B.D. and Scott, M.J. (2009). Effectiveness of using a video game to teach a course in mechanical engineering. *Computers & Education* 53 (3): 900–912.

Conati, C. and Kardan, S. (2013). Student modeling: supporting personalized instruction, from problem solving to exploratory open-ended activities. *AI Magazine* 34 (3): 13–26.

Conati, C., Porayska-Pomsta, K., and Mavrikis, M. (2018). AI in education needs interpretable machine learning: Lessons from open learner modelling. *arXiv*. https://doi.org/10.48550/arXiv.1807.00154.

D'Mello, S. and Graesser, A. (2015). Feeling, thinking, and computing with affect-aware learning technologies. In: *The Oxford Handbook of Affective Computing*, (ed. R.A. Calvo, S. D'Mello, J. Gratch, and A. Kappas), (pp. 419–434). Oxford University Press.

D'Mello, S., Dieterle, E., and Duckworth, A. (2017). Advanced, analytic, automated (AAA) measurement of engagement during learning. *Educational Psychologist* 52 (2): 104–123.

de Jong, T., Lazonder, A.W., Chinn, C.A., et al. (2023). Let's talk evidence - The case for combining inquiry-based and direct instruction. *Educational Research Review* 39: 100536. https://doi.org/10.1016/j.edurev.2023.100536.

Deshpande, A.A. and Huang, S.H. (2011). Simulation games in engineering education: a state–of–the–art review. *Computer Applications in Engineering Education* 19 (3): 399–410.

Dever, D., Wiedbusch, M., Marano, C. et al. (2024). Product and process: Game mechanics for science learning. *International Journal of Serious Games* 11: 127–153.

Dever, D., Wiedbusch, M., Romero, S. et al. (in press). Investigating pedagogical agents' scaffolding of self-regulated learning in relation to learners' subgoals. *British Journal of Educational Technology*.

Dever, D.A., Amon, M.J., Vrzáková, H., et al. (2022) Capturing sequences of learners' self-regulatory interactions with instructional material during game-based learning using auto-recurrence quantification analysis. *Frontiers in Psychology* 13: 813677. https://doi.org/10.3389/fpsyg.2022.813677.

Dever, D.A., Sonnenfeld, N.A., Wiedbusch, M.D., et al. (2023a). A complex systems approach to analyzing pedagogical agents' scaffolding of self-regulated learning within an intelligent tutoring system. *Metacognition and Learning* 18 (3): 659–691. https://doi.org/10.1007/s11409-023-09346-x.

Dever, D.A., Wiedbusch, M.D., Romero, S.M., et al. (2023b). Identifying the effects of scaffolding on learners' temporal deployment of self-regulated learning operations during game-based learning using multimodal data. *Frontiers in Psychology* 14: 1280566.

duBoulay, B., Mitrovic, T., and Yacef, K. (2023) (Eds.). *Handbook of Artificial Intelligence in Education*. Edward Elger Publishing.

Dym, C.L., Agogino, A.M., Eris, O., et al. (2005). Engineering design thinking, teaching, and learning. *Journal of Engineering Education* 94 (1): 103–120.

Ebner, M. and Holzinger, A. (2007). Successful implementation of user-centered game based learning in higher education: an example from civil engineering. *Computers & Education* 49 (3): 873–890.

Ericsson, K.A. and Simon, H.A. (1993). *Protocol Analysis: Verbal Reports as Data* (Rev. ed.). MIT Press.

Geisinger, B.N. and Raman, D.R. (2013). *Why They Leave: Understanding Student Attrition from Engineering Majors*. Iowa State University Digital Repository.

Giannakos, M., Spikol, D., Di Mitri, D. et al. (2022). Introduction to multimodal learning analytics. In: *The Multimodal Learning Analytics Handbook* (ed. M. Giannakos et al.). Springer. https://doi.org/10.1007/978-3-031-08076-0_1.

Gong, Y., Guo, J., Li, J. et al. (2019). Experimental realization of an intrinsic magnetic topological insulator. *Chinese Physics Letters* 36 (7): 076801. https://doi.org/10.1088/0256-307X/36/7/076801.

Greene, J.A. and Azevedo, R. (2009). A macro-level analysis of SRL processes and their relations to the acquisition of sophisticated mental models. *Contemporary Educational Psychology* 34 (1): 18–29. https://doi.org/10.1016/j.cedpsych.2008.05.006.

Greene, J.A., Deekens, V., Copeland, D., et al. (2018). Capturing and modeling self-regulated learning using think-aloud protocols. In: *Handbook of Self-regulation of Learning and Performance*, 2e, (ed. D. Schunk and J.A. Greene) (2nd ed., pp. 323–337). Routledge.

Halszka, J., Holmqvist, K., and Gruber, H. (2017). Eye tracking in educational science: theoretical frameworks and research agendas. *Journal of Eye Movement Research* 10 (1): 1–18.

Harley, J.M., Bouchet, F., Hussain, S., et al. (2015). A multi-componential analysis of emotions during complex learning with an intelligent multi-agent system. *Computers in Human Behavior* 48: 615–625. https://doi.org/10.1016/j.chb.2015.02.013.

Harley, J.M., Lajoie, S.P., Frasson, C. et al. (2017). Developing emotion-aware, advanced learning technologies: A taxonomy of approaches and features. *International Journal of Artificial Intelligence in Education* 27: 268–297. https://doi.org/10.1007/s40593-016-0126-8.

Harley, J.M., Lou, N.M., Liu, Y. et al. (2021). University students' negative emotions in a computer-based examination: The roles of trait test-emotion, prior test-taking methods, and gender. *Assessment and Evaluation in Higher Education* 46 (6): 956–972. https://doi.org/10.1080/02602938.2020.1836123.

Harley, J.M., Pekrun, R., Taxer, J.L. et al. (2019). Emotion regulation in achievement situations: An integrated model. *Educational Psychologist* 54 (2): 106–126. https://doi.org/10.1080/00461520.2019.1587297.

Hattie, J. and Timperley, H. (2007). The power of feedback. *Review of Educational Research* 77 (1): 81–112.

Hwang, G.J., Sung, H.Y., Hung, C.M., et al. (2017). Development of a personalized educational computer game based on students' learning styles. *Educational Technology Research and Development* 65 (1): 327–351.

Hwang, T.J., Rabheru, K., Peisah, C. et al. (2020). Loneliness and social isolation during the COVID-19 pandemic. *International Psychogeriatrics* 32 (10): 1217–1220. https://doi.org/10.1017/S1041610220000988.

Järvelä, S. and Bannert, M. (2021) Temporal and adaptive processes of regulated learning: what can multimodal data tell? *Learning and Instruction* 72: 1011268. https://doi.org/10.1016/j.learninstruc.2019.101268.

Jonassen, D., Strobel, J., and Lee, C.B. (2006). Everyday problem solving in engineering: Lessons for engineering educators. *Journal of Engineering Education* 95 (2): 139–151.

Karabenick, S.A. and Berger, J.L. (2013). Help seeking as a self-regulated learning strategy. In: *Applications of Self-regulated Learning Across Diverse Disciplines: A Tribute to Barry J. Zimmerman*, (ed. H. Bembenutty, T.J. Cleary, and A. Kitsantas), (pp. 237–261). Information Age Publishing.

Ke, F. (2009). A qualitative meta-analysis of computer games as learning tools. In R.E. Ferdig (Ed.), *Handbook of Research on Effective Electronic Gaming in Education* (pp. 1–32). IGI Global.

Kickmeier-Rust, M.D. and Albert, D. (2010). Micro-adaptivity: Protecting immersion in didactically adaptive digital educational games. *Journal of Computer Assisted Learning* 26 (2): 95–105.

Kober, N. (2014). *Reaching Students: What Research Says About Effective Instruction in Undergraduate Science and Engineering* (1st ed.). National Academies Press. https://doi.org/10.17226/18687.

Koedinger, K.R., Brunskill, E., Baker, R.S., et al. (2013). New potentials for data-driven intelligent tutoring system development and optimization. *AI Magazine* 34 (3): 27–41.

Koskinen, A., McMullen, J., Hannula-Sormunen, M., et al. (2023). The strength and direction of the difficulty adaptation affect situational interest in game-based learning. *Computers & Education* 194: 104694.

Kostons, D., van Gog, T., and Paas, F. (2012). Training self-assessment and task-selection skills: a cognitive approach to improving self-regulated learning. *Learning and Instruction* 22 (2): 121–132.

Kovanovic, V., Azevedo, R., Gibson, D. et al. (Eds.) (2023). *Unobtrusive Observations of Learning in Digital Environments: Examining Behaviors, Cognition, Emotion, Metacognition, and Social Processes Using Learning Analytics*. Springer.

Lamb, R.L., Annetta, L., Firestone, J., et al. (2018). A meta-analysis with examination of moderators of student cognition, affect, and learning out-

comes while using serious educational games, serious games, and simulations. *Computers in Human Behavior* 80: 158–167.

Lester, J., Bansal, M., Biswas, G., et al. (2024). The AI Institute for Engaged Learning. *AI Magazine* 45 (1): 69–76. https://doi.org/10.1002/aaai.12161.

Lin, X., Hmelo, C., Kinzer, C.K., et al. (1999). Designing technology to support reflection. *Educational Technology Research and Development* 47 (3): 43–62.

Locke, E.A. and Latham, G.P. (2002). Building a practically useful theory of goal setting and task motivation: a 35-year odyssey. *American Psychologist* 57 (9): 705–717.

Lopes, R. and Bidarra, R. (2011). Adaptivity challenges in games and simulations: a survey. *IEEE Transactions on Computational Intelligence and AI in Games* 3 (2): 85–99.

Mayer, R. (2019). Thirty years of research on online research. *Applied Cognitive Psychology* 33: 152–159.

Molenaar, I. (2021). Personalization of learning: Towards hybrid human-AI learning technologies. In: *OECD Digital Education Outlook 2021: Pushing the Frontiers with AI, Blockchain, and Robots* (pp. 57–77). OECD Publishing. https://doi.org/10.1787/589b283f-en.

Molenaar, I. and Järvelä, S. (2014). Sequential and temporal characteristics of self and socially regulated learning. *Metacognition and Learning* 9 (2): 75–85.

Molenaar, I. and Knoop-van Campen, C.A.N. (2019). How teachers make dashboard information actionable. *IEEE Transactions on Learning Technologies* 12 (3): 347–355.

Molenaar, I., de Mooij, S., Azevedo, et al. (2023). Measuring self-regulated learning and the role of AI: five years of research using multimodal multichannel data. *Computers in Human Behavior* 139: 107540. https://doi.org/10.1016/j.chb.2022.107540.

National Academies of Sciences, Engineering, and Medicine. (2018). *How People Learn II: Learners, Contexts, and Cultures*. The National Academies Press. https://doi.org/10.17226/24783.

National Academy of Engineering. (2013). *The future engineering: Annual Report*. National Academies of Science, Engineering, and Medicine.

National Academy of Engineering. (2018). *Understanding the educational and career pathways of engineers*. The National Academies Press. https://doi.org/10.17226/25284.

Nguyen, A., Jarvela, S., Rose, C. et al. (2023). Examining socially shared regulation and shared physiological arousal events with multimodal learning analytics. *British Journal of Educational Technology* 54 (1): 293–312. https://doi.org/10.1111/bjet.13280.

Ninaus, M. and Nebel, S. (2021). A systematic literature review of analytics for adaptivity within educational video games. *Frontiers in Education* 5: 611072. https://doi.org/10.3389/feduc.2020.611072.

Pawley, A.L. (2009). Universalized narratives: Patterns in how faculty members define "engineering". *Journal of Engineering Education* 98 (4): 309–319.

Pintrich, P.R. (2000). The role of goal orientation in self-regulated learning. In: *Handbook of self-regulation* (ed. M. Boekaerts, P.R. Pintrich, and M. Zeidner), (pp. 451–502). Academic.

Plass, J., Horner, B., and Mayer, R. (Eds.) (2019). *Handbook of Game-based Learning*. MIT Press.

Plass, J.L. and Pawar, S. (2020). Toward a taxonomy of adaptivity for learning. *Journal of Research on Technology in Education* 52 (3): 275–300.

Rahimi, F., Boyd, J., Eiserman, J. et al. (2022). Museum beyond physical walls: An exploration of virtual reality-enhanced experience in an exhibition-like space. *Virtual Reality* 26: 1471–1488. https://doi.org/10.1007/s10055-022-00643-5.

Roll, I. and Winne, P.H. (2015). Understanding, evaluating, and supporting self-regulated learning using learning analytics. *Journal of Learning Analytics* 2 (1): 7–12. https://doi.org/10.18608/jla.2015.21.2.

Ruipérez-Valiente, J.A., Muñoz-Merino, P.J., and Delgado Kloos, C. (2017). Detecting and clustering students by their gamification behavior with badges: A case study in engineering education. *International Journal of Engineering Education* 33 (2): 816–830.

Sailer, M. and Homner, L. (2020). The gamification of learning: A meta-analysis. *Educational Psychology Review* 32 (1): 77–112. https://doi.org/10.1007/s10648-019-09498-w.

Sampayo-Vargas, S., Cope, C.J., He, Z., et al. (2013). The effectiveness of adaptive difficulty adjustments on students' motivation and learning in an educational computer game. *Computers & Education* 69: 452–462.

Sawyer, R.K. and Henriksen, D. (2023). *Explaining Creativity: The Science of Human Innovation* (3rd ed.). Oxford University Press.

https://doi.org/10.1093/oso/9780197747537.001.0001.

Shute, V. and Ventura, M. (2017). Stealth assessment. In: *The SAGE Encyclopedia of Educational Technology* (pp. 675–676). SAGE.

Shute, V.J. (2023). History of stealth assessment and a peek into its future. In: *Games as Stealth Assessments* (ed. M.P. McCreery and S.K. Krach) (pp. 1–23). IGI Global. https://doi.org/10.4018/979-8-3693-0568-3.ch001.

Shute, V.J. and Rahimi, S. (2022). Stealth assessment: A primer. In: *Rapid Community Report Series* (pp. 1–11). Digital Promise and the International Society of the Learning Sciences. https://repository.isls.org//handle/1/7671.

Spikol, D., Ruffaldi, E., Dabisias, G., et al. (2018). Supervised machine learning in multimodal learning analytics for estimating success in project-based learning. *Journal of Computer Assisted Learning* 34 (4): 366–377.

Taub, M., Azevedo, R., Bouchet, F., et al. (2014). Can the use of cognitive and metacognitive self-regulated learning strategies be predicted by learners' levels of prior knowledge in hypermedia-learning environments? *Computers in Human Behavior* 39: 356–367.

Taub, M., Azevedo, R., Rajendran, R. et al. (2021). How are students' emotions related to the accuracy of cognitive and metacognitive processes during learning with an intelligent tutoring system? *Learning and Instruction* 72. https://doi.org/10.1016/j.learninstruc.2019.04.001.

Taub, M., Mudrick., N., Bradbury, A.E. et al. (2019). Self-regulation, self-explanation, and reflection in game-based learning. In: *Handbook of Game-based Learning* (ed. J. Plass, B. Horner, and R. Mayer) (pp. 239–262). MIT Press.

Tsai, M.J., Huang, L.J., Hou, H.T., et al. (2019). Visual behavior, flow and achievement in game-based learning. *Computers & Education* 98: 115–129.

Tsai, Y.L. and Tsai, C.C. (2020). A meta-analysis of research on digital game-based learning in science education. *Computers & Education* 157: 103994.

Tseng, M.L., Chang, C.H., Lin, C.W.R. et al. (2020). Future trends and guidance for the triple bottom line and sustainability: A data-driven bibliometric analysis. *Environmental Science and Pollution Research* 27: 33543–33567. https://doi.org/10.1007/s11356-020-09284-0.

Turns, J.A., Sattler, B., Yasuhara, K., et al. (2014). Integrating reflection into engineering education. In: *Proceedings of the ASEE Annual Conference and Exposition* (pp. 24–1286).

Vandewaetere, M., Desmet, P., and Clarebout, G. (2011). The contribution of learner characteristics in the development of computer-based adaptive learning environments. *Computers in Human Behavior* 27 (1): 118–130.

VanLehn, K. (2011). The relative effectiveness of human tutoring, intelligent tutoring systems, and other tutoring systems. *Educational Psychologist* 46 (4): 197–221.

Vlachopoulos, D. and Makri, A. (2017). The effect of games and simulations on higher education: A systematic literature review. *International Journal of Educational Technology in Higher Education* 14 (1): 1–33. https://doi.org/10.1186/s41239-017-0062-1.

Walkington, C. and Bernacki, M.L. (2020). Appraising research on personalized learning: definitions, theoretical alignment, advancements, and future directions. *Journal of Research on Technology in Education* 52 (3): 235–252.

Wiedbusch, M., Azevedo, R., Chi, M. et al. (2021). Intelligent teacher dashboards that support students' self-regulated learning, engagement, and teachers' decision making. *Frontiers in Education* 6. https://doi.org/10.3389/feduc.2021.570229.

Wiedbusch, M., Brosnihan, A., Delgado, T. et al. (2024). From game-based simulation to practice: The challenges of capturing, modeling, and transferring multimodal data for chemistry skill mastery. In: *Proceedings of the 10th Joint International Conference on Serious Games* (ed. J. Plass and X. Ochoa) (pp. 447–454). Springer.

Wiedbusch, M., Dever, D., Li, S. et al. (2023). Measuring multidimensional facets of SRL engagement with multimodal data. In: *Unobtrusive measures of engagement* (ed. D. Ifanthaler, R. Azevedo, Gibson, and V. Kovanonic) (pp. 141–173). Springer.

Wiedbusch, M.D., Dever, D.A., Wortha, F. et al. (2021). Revealing data feature differences between system and learner-initiated self-regulated learning processes within hypermedia through principal component analysis. In: *HCII'21: Proceedings from the Adaptive Instructional Systems. Design and Evaluation, Lecture Notes in Computer Science* (ed. R.A. Sottilare and J. Schwarz) (vol. 12792, pp. 481–495). Springer.

Winne, P.H. (2018). Cognition and metacognition within self-regulated learning. In: *Handbook of Self-regulation of Learning and Performance*, 2e (ed. D. Schunk and J. Greene) (pp. 36–48). Routledge.

Winne, P.H. and Azevedo, R. (2022). Metacognition. In: K. Sawyer (Ed.), *Cambridge handbook of the learning sciences* (3rd ed., pp. 63–87). Cambridge University Press.

Winne, P.H. and Hadwin, A. (2011). *nStudy: A web application for researching and promoting self-regulated learning* (version 2.1) [computer program]. Simon Fraser University.

Winne, P.H. and Hadwin, A.F. (1998). Studying as self-regulated learning. In: *Metacognition in educational theory and practice* (ed. D.J. Hacker, J. Dunlosky, and A.C. Graesser) (pp. 277–304). Routledge. https://doi.org/10.4324/9781410602350.

Winne, P.H. and Hadwin, A.F. (2013). nStudy: tracing and supporting self-regulated learning in the internet. In: *International Handbook of Metacognition and Learning Technologies*, (ed. R. Azevedo and V. Aleven), (pp. 293–308). Springer.

Worsley, M. and Blikstein, P. (2018). A multimodal analysis of making. *International Journal of Artificial Intelligence in Education* 28 (3): 385–419. https://doi.org/10.1007/s40593-017-0160-1.

Zimmerman, B.J. (2002). Becoming a self-regulated learner: an overview. *Theory into Practice* 41 (2): 64–70.

Zimmerman, B.J. and Moylan, A.R. (2009). Self-regulation: where metacognition and motivation intersect. In: *Handbook of Metacognition in Education*, (ed. D.J. Hacker, J. Dunlosky, and A.C. Graesser), (pp. 299–315). Routledge.